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CS 499 - 5-1 Journal

Professor Phillips

February 8, 2026

Part One:

One significant trend in computer science is the evolution of Artificial Intelligence (AI) to enhance game intelligence and adaptive gameplay. According to Bradshaw (2023), AI is moving beyond simple scripted movements to create characters that reason logically and adjust to players actions in real time. This is transformative because it shifts software development from static loops to dynamic, generative environments where the experience evolves based on user input. For workers and consumers, this means more engaging simulations and more effective training tools that mirror the unpredictability of the real world. Personally, this fits my interest in software engineering as it challenges me to build more sophisticated logic and modular code. Through the successful completion of my ePortfolio enhancements, I have now achieved all the course outcomes discussed in my initial plan, including designing computing solutions with algorithmic trade offs (Outcome 3), implementing industry standard technical tools (Outcome 4), and applying a security mindset to mitigate design flaws (Outcome 5).

A second major development is the advancement of Virtual Reality (VR) and Simultaneous Localization and Mapping (SLAM). Matthews (2018) explains that SLAM technology allows for the instant translation of real world data into virtual environments using sensors. This technology is changing the field by moving computing away from 2D screens and into 3D physical spaces, which is critical for fields like medical training and autonomous vehicle navigation. This trend improves the experience of citizens and workers by providing safe, high

fidelity environments for practicing delicate surgeries or flight maneuvers without real world risk. Professionally, I am interested in how these immersive systems handle massive amounts of spatial data.

Part Two: Progress Update

Checkpoint	Software Design and Engineering	Algorithms and Data Structures	Databases
Name of Artifact Used	Weight Tracking App (CS 360)	Binary Search Tree (CS 300)	Animal Shelter Project (CS 340)
Status of Initial Enhancement	Completed. Refactored code to the MVVM design pattern with LiveData.	Completed. Converted BST to a self-balancing AVL Tree.	Completed. Integrated MongoDB with a Python based CRUD dashboard.
Submission Status	Milestone Two (Enhancement One) submitted.	Milestone Three (Enhancement Two) submitted.	Milestone Four (Enhancement Three) submitted.
Status of Final Enhancement	Functionality is stable; architecture is fully modular.	Optimized memory and interactive search.	Implemented RESTful Web API using Flask.
Uploaded to ePortfolio	Code uploaded to GitHub; narrative finished.	Code uploaded to GitHub; narrative finished.	Code uploaded to GitHub; narrative finished.
Status of Finalized ePortfolio	Ready for instructor review.	Ready for instructor review.	Ready for instructor review.

References

Bradshaw, R. (2023, June 30). *The Future of Video Gaming: Technologies That*

Will Transform Our Experiences. Apollo Technical LLC.

<https://www.apollotechnical.com/the-future-of-video-gaming/>

Matthews, P. (2018, May 1). *The future of VR and Ar*. IEEE Computer Society.

<https://www.computer.org/publications/tech-news/trends/the-future-of-vr-a>

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